Math 242, S15 **Name:** Answer

Quiz 12 **SSN:** xxx-xx-

<u>Instructions:</u> **There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.**

Problem 1. (6 points) Use Laplace transform to solve the initial value problem

$$x'' + 9x = 4t; \quad x(0) = x'(0) = 0.$$

Solution: let $\mathcal{L}\{x(t)\} = X(s)$, then

$\mathcal{L}\{x'(t)\} = sX(s)$, $\mathcal{L}\{x''(t)\} = s^2X(s)$

Thus

$\mathcal{L}\{x''+9x\} = \mathcal{L}\{4t\}$

$\mathcal{L}\{x''\} + 9\mathcal{L}\{x\} = 4 \cdot \frac{1}{s^2}$

$s^2X(s) + 9X(s) = \frac{4}{s^2}$

$(s^2+9)X(s) = \frac{4}{s^2}$

$X(s) = \frac{4}{s^2(s^2+9)}$

Check

$\frac{4}{s^2(s^2+9)} = \frac{A}{s^2} + \frac{B}{s^2+9} = \frac{A(s^2+9)+Bs^2}{s^2(s^2+9)}$

$= \frac{(A+B)s^2+9A}{s^2(s^2+9)}$

$\Rightarrow \begin{cases} A+B=0 \\ 9A=4 \end{cases} \Rightarrow \begin{matrix} A = \frac{4}{9} \\ B = -\frac{4}{9} \end{matrix}$

Thus

$x(t) = \mathcal{L}^{-1}\left\{\frac{4}{s^2(s^2+9)}\right\}$

$= \mathcal{L}^{-1}\left\{\frac{4}{9}\frac{1}{s^2} - \frac{4}{9}\frac{1}{s^2+9}\right\}$

$= \frac{4}{9}t - \frac{4}{27}\sin 3t$

Problem 2. (4 points) Use **Theorem of Differential Tansforms** (i.e., $\mathcal{L}\{tf(t)\} = -F'(s)$ where $F(s) = \mathcal{L}\{f(t)\}$) to compute $\mathcal{L}\{t\cos 3t\}$.

Solution: let $f(t) = \cos 3t \Rightarrow F(s) = \mathcal{L}\{\cos 3t\} = \frac{s}{s^2+9}$

Thus $\mathcal{L}\{t\cos 3t\} = -\frac{d}{ds}\left(\frac{s}{s^2+9}\right)$

$= -\left[\frac{(s^2+9)-s(2s)}{(s^2+9)^2}\right]$

$= -\frac{9-s^2}{(s^2+9)^2} = \frac{s^2-9}{(s^2+9)^2}$